Campus ICT Fault Reporting and Support System

MINI USER MANUAL | Java Swing desktop application

Use this system to register student/staff reporters, manage ICT technicians, report campus ICT faults, and track their assignment and resolution. Records are stored in the MySQL database **campus_ict_db**.

1 First-time setup and startup

Requirements: Java 17 or later and MySQL Server. Extract the submission ZIP before running. Keep Executable, Database, Source_code, and user_manual together. NetBeans and Maven are not needed to run the packaged application.

1. Start the MySQL service. On Windows, press **Win+R**, enter **services.msc**, and start the appropriate MySQL service if stopped.

2. On a new computer, open MySQL Workbench and connect. Select **File > Open SQL Script**, open **Database/campus_ict_db.sql** in the extracted submission, review it, and execute it to create/import the desktop database.

Import caution: The export includes DROP TABLE statements and replaces the listed tables and data. Back up an existing database first. If campus_ict_db already contains the correct tables and records, skip importing it again.

3. In **Executable/config**, copy **db.properties.example** and rename the copy **db.properties**. Edit it in Notepad with the local connection details:

```
db.url=jdbc:mysql://127.0.0.1:3307/campus_ict_db?useSSL=false&serverTimezone=UTC
db.username=YOUR_MYSQL_USERNAME
db.password=YOUR_MYSQL_PASSWORD
```

Replace both placeholders. Port **3307** is the development-machine setting; use the actual server host/port on another computer. Save the file. Keep real credentials private.

4. Open **Executable** and double-click **RunCampusICT.cmd**. The launcher opens the Java Main Menu. For later use, start MySQL and run this launcher; do not re-import the database each time.

PowerShell alternative: Open a terminal in the Executable folder and run:

```
.\RunCampusICT.cmd
```

2 Navigate the five forms

Form	Purpose
Main Menu	Open the four functional forms or exit the application.
Users Management	Register and maintain Student/Staff reporter details.
Technicians Management	Maintain technician details, department, and availability.
Report Fault	Submit a fault report; select an existing report to update or delete.
Search & Manage Tickets	Find tickets, edit details, assign technicians, and update resolution status.

Navigation: Use **Back to Main Menu** to switch forms. **Clear** resets inputs and selection without deleting data. **Refresh** reloads saved records.

Working with records

Register reporters first, then submit faults and manage support.

3 Add and maintain users or technicians

1. Open **Users Management**. Enter Full Name, a unique Email, optional Phone, and Role (**Student** or **Staff**). Click **Add**; MySQL generates the User ID.

2. Open **Technicians Management**. Enter Technician Name, Department, unique Email, and Status (**Available**, **Busy**, or **On Leave**). Click **Add**.

3. Use the search box to find users/technicians by ID, name, or email. Select a table row, change the required details, then click **Update**. Check the refreshed table.

4 Report a fault

1. Open **Report Fault** and select the registered reporter from the dropdown.

2. Choose a Category: **Wi-Fi / Network, Hardware, Software, Lab Equipment,** or **Other**. Enter the campus Location and a clear Description.

3. Choose Urgency: **Low, Medium, High,** or **Critical**. Click **Submit Fault**. Note the generated Ticket ID. New tickets start **Pending** with no technician assigned.

4. To edit a report, select its table row, change the report fields, and click **Update**. This edits the existing report; use Submit Fault to create a new one.

5 Search, assign, and resolve tickets

1. Open **Search & Manage Tickets**. Use the available search/filter controls to locate a ticket, then select its table row.

2. Select an appropriate technician, set the status to **In Progress**, and click **Update** to save the assignment.

3. When the fault is fixed, enter the resolution notes, choose **Resolved**, and click **Update**. Check that the saved status and assignment appear in the table.

6 Delete records and handle common issues

Delete: Select a table row in Users Management, Technicians Management, Report Fault, or Search & Manage Tickets. Click **Delete** and read the confirmation. Choose **No** to cancel or **Yes** to permanently remove the selected record. Use disposable data when practising.

Linked records: A user with reported tickets cannot be deleted. Deleting a technician keeps their tickets but clears the assignment. Deleting a ticket does not delete its reporter or technician.

Missing/old data: Click Refresh or reopen the form. Both ticket forms share the same MySQL records; an already displayed table does not refresh instantly.

Validation: Complete required fields, use a valid unique email, and keep Location within 100 characters. A reporter must exist before a fault can be submitted.

Connection failure: Check the MySQL service, port, database name, and config/db.properties. **Exit:** Return to Main Menu and click Exit.